

MEMORIA: A Memory Enhancement and MOment Retrleval Application at the LSC2025

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Abstract

The documentation of everyday life has become increasingly popular with the rise of wearable devices, leading to a higher demand for systems that can process, store, and organize this type of data called lifelogs. This paper presents the latest version of MEMORIA, built to participate in the eighth edition of the Lifelog Search Challenge. The enhancements of this year include improved annotation pipelines by updating models, the addition of BLIP (Bootstrapped Language-Image Pretraining), image embeddings with Milvus, and the introduction of an automatic query parser leveraging NLP techniques and a Large Language Model. These enhancements allowed MEMORIA to get the best overall score in the LSC.

CCS Concepts

• Information systems → Search interfaces; • Human-centered computing → Interactive systems and tools.

Keywords

lifelog, lifelogging, image processing, image annotation, data retrieval, object detection, Machine Learning, Information Systems, Large Language Models

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Team Name

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1 Introduction

Lifelogging is the automated and continuous recording of personal experiences using the ever-increasing diversity of wearable devices, cameras, smartphones, and other sensors. Each piece of collected data, whether images, videos, biometric signals, location traces, or audio, constitutes a lifelog. These multimodal datasets encapsulate

the rich aspects of daily life, offering unprecedented opportunities for memory enhancement, health monitoring, and behavioral analysis. However, the sheer volume, heterogeneity, and unstructured nature of lifelog data pose significant challenges for efficient retrieval and organization. As lifelogs rapidly accumulate into vast archives, systems capable of analyzing, structuring, storing, and retrieving specific moments in lifelog data became increasingly popular.

The Lifelog Search Challenge (LSC) is an annual event that serves as a benchmark competition where researchers can evaluate their retrieval systems via different tasks. This year, the Lifelog Search Challenge 2025 [9], its 8th edition, will once again provide three different tasks and a shared dataset to be used by the various systems. The Known-Item Search (KIS) provides a text query that represents a single moment in the lifelog data, over time the query gets more and more detailed, and the teams must provide a single image that is part of the moment. The Ad-hoc Search provides a topic that usually represents an activity, moment, or event that can happen numerous times in the data set and requires teams to provide as many examples of that as they can. The last task, the QA task, provides a topic in which teams must provide a textual answer to it.

This paper presents the latest version of MEMORIA, or Memory Enhancement and MOment Retrleval Application. This year's version uses the improved interface from last year [8], with improvements on the query parsing by using a large language model to detect keywords and misspells, and a similar retrieval system to last year, but with updated annotations and performance improvements. There is also ongoing work on a vector search system leveraging BLIP2 embeddings and Milvus¹.

2 Related Work

With the increase in popularity of lifelogging systems, more and more lifelog systems are being researched and developed. The eighth iteration of the Lifelog Search Challenge (LSC25) aims to test the performance and usability of different systems by putting them through different types of tasks.

LifeXplore [21] is a lifelog system that allows for text search using mainly FAISS [14] and OpenCLIP [4] as default. It additionally allows the creation of queries that combine the default search with other modalities such as semantic concepts using EfficientNet B2 [22], objects using YOLOv7 [25], and recognized text using CRAFT [3].

¹<https://milvus.io>



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SnapSeek [11] is another lifelogging system that allows image search using textual queries. The system combines two search approaches. The first uses the embeddings generated from OpenCLIP [4], BLIP2 [16], BEiT3 [28], and all-mpnetbase-v2 connected to a Milvus database. The second instead uses the embedding generated from PaddleOCR [6] (optical character recognition), YOLOv8 and YOLOv9 [13][26] (object detection), GIT [27] (image captioning) and ViT-L/14-336 [30] (activity detection) connected to an Elasticsearch engine. It has several key functionalities, namely, metadata enhancement, standardization of queries, information auto-parsing, an effective integrative ranking system, and an intuitive user interface.

VISIONE [2] is a video retrieval system designed to analyze, index, and search for video content. Some adjustments were made to allow and facilitate image retrieval tasks. It has four main search functionalities: text-to-image search, similarity search by searching for images similar to an uploaded one, object and color-based search, and temporal search. It uses three different multimodal feature extractors to enhance search capabilities: the e OpenCLIP ViT-L/14 model named ClipLAION [20] trained on the LAION-2B dataset, the CLIP2Video [7] and ALADIN [17]. The system employs two indexing and searching methods: the Facebook FAISS ² library to store and access the CLIP2Video and ClipLAION features, and Apache Lucene ³ to store all the other data.

Libro [10] is another video retrieval system adapted for image retrieval tasks. The system treats the camera images like frames of a video with the images of one day being treated as a single video. Libro offers image-to-text and text-to-image search capabilities. All the images are pre-processed into keyframes and shots to reduce duplicates and user confusion.

MemoriEase [23] provides a conversational search interface for users to search for images with assistance from an AI assistant to provide a more user-friendly interface for users to explore. The system offers data processing using BLIP2’s generated embeddings from images, data indexing into an Elasticsearch engine, and a chat interface with an OpenAI API. The system also uses Retrieval-Augmented Generation (RAG) to help the LLM used in the chat interface in answering questions made by the user, using the lifelog data of the images in the album.

Lifeinsight [24] is a lifelog retrieval system that leverages the strength of BLIP and CLIP for image annotation. The new version allows for temporal information in the queries while providing an automatic query parser that extracts key concepts like main context, location, date, and time from the user query.

Memento [1] organizes lifelog data into events enabling a moment-centered retrieval over an image-centered retrieval. It also incorporates a question-answer pipeline using GPT-3.5 Turbo and Mistral7B to allow users to use natural language when retrieving lifelog information. The system uses CLIP embeddings to discover similarities between adjacent images and groups them based on such embeddings into events. Image annotation is done using BLIP2 which creates a caption for each image. Using the captions of each image inside an event, two different summaries for each event are made,

one with GPT-3.5 Turbo, and another with Mistral7B. Using the LlamaIndex framework, a Retrieval Augmentation Generation (RAG) pipeline allows for question-answering.

3 MEMORIA System

MEMORIA is a web-based platform designed for storing, organizing, analyzing, visualizing, and retrieving lifelog data. It integrates a collection of modular components, each tailored to support and enhance different aspects of the lifelogging experience. These modules handle data storage, organization, automated annotation, retrieval, and visualization tasks.

An overview of the projected MEMORIA architecture for LSC25 is presented in Figure 1, illustrating the connection between core components.

This section offers an overview of MEMORIA’s main modules, outlining their roles and how they contribute to the system’s functionality.

3.1 Image Annotation

MEMORIA keeps its core idea of generating textual annotations for each image and storing them for further retrieval. This year, the models used for annotations were updated and MEMORIA generates its annotations based on five categories:

- **Object Detection:** Previously, MEMORIA handled the detection of objects via YOLOv7 and GRiT [29], but after use in the LSC, we deemed that YOLO annotations were enough to have a good understanding of the objects in the images. After analyzing the work of Khanam et al. [15], we decided to upgrade to YOLOv11 ⁴ and use it as the sole source of object annotations.
- **Text Recognition:** Previously, MEMORIA used CRAFT for text area recognition and OCR, but this year, we switched to PaddleOCR as we deemed it to be more accurate in certain scenarios.
- **Image Captioning:** For the generation of image captions, MEMORIA still uses ClipCap [18], but BLIP2-opt-2.7b was also added to generate captions. We found that often times these captions complement each other and facilitate the retrieval process.
- **Scene Recognition:** For scene recognition MEMORIA keeps the same Resnet18 model trained on the Places365 dataset [31] and it provides valuable information on the environment the images take place.
- **Metadata:** Location data and time from the images are also used and stored so they can be used as filters when querying for when or where a specific action took place.

3.2 Image Embeddings

Motivated by the work of the SnapSeek team at LSC24 [11] and the knowledge that converting image embeddings to text for retrieval inherently loses information, in this work, MEMORIA generates image embeddings for all dataset images using clip-vit-base-patch32.

²<https://github.com/facebookresearch/faiss>

³<https://lucene.apache.org/>

⁴<https://docs.ultralytics.com/models/yolo11/>

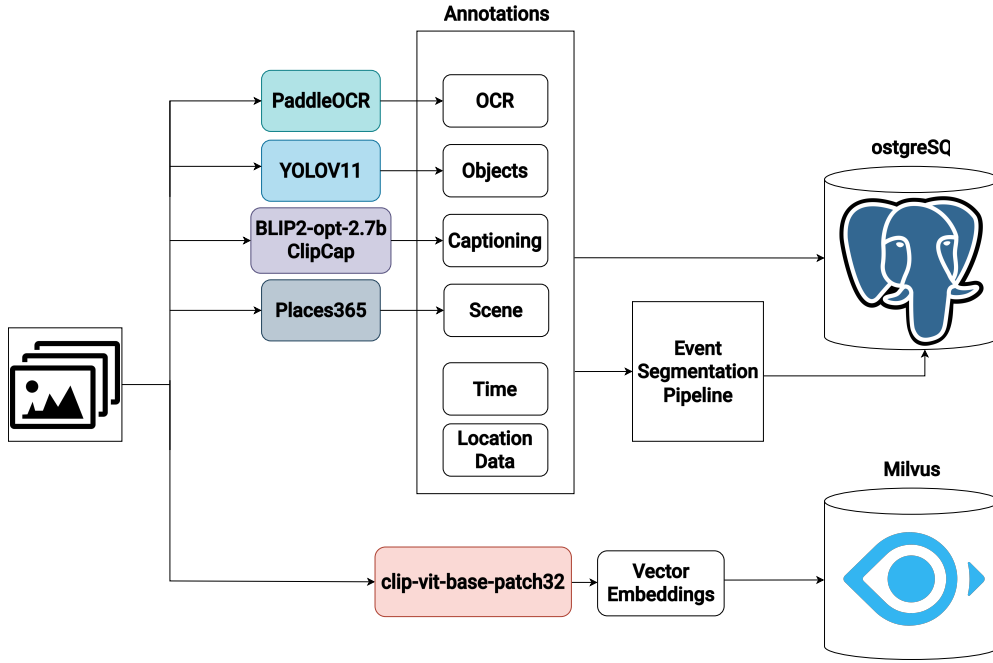


Figure 1: Overview of the projected MEMORIA lifelog system architecture. The system comprises interconnected blocks representing the key components described in this paper.

These embeddings are stored in a Milvus vector collection. The introduction of Milvus allowed MEMORIA’s retrieval to be enhanced, both the text query retrieval and the image to image retrieval.

3.3 Event Segmentation

Event segmentation can be a powerful tool in lifelog organization by dividing moments into coherent temporal segments, each corresponding to a distinct event. This structuring can not only improve the efficiency of search and navigation but also align better with how humans recall memories through meaningful episodes rather than isolated images. In addition, segmenting lifelogs into events improves the system interface by reducing visual clutter and conserving screen space.

MEMORIA retains the hierarchical event segmentation framework introduced in the previous version [8], where it clusters images into blocks called events. It consists of five hierarchical layers: days, parts of the day, location, environment, and image similarity. These events are constructed using preprocessed metadata and annotation results and incorporate temporal, locational, environmental, visual, and semantic information.

The new feature this year’s version brings into the event segmentation pipeline is the addition of an event caption or description that describes what is happening during a certain event. This description is generated using Mistral7B LLM [12] and uses the image captions and location data of the events.

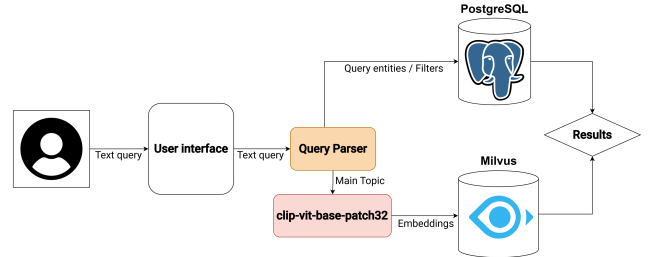


Figure 2: Overview of the process of using a text query for image retrieval in MEMORIA.

3.4 Retrieval

The system offers three different ways to search for lifelog data. These methods offer both flexibility and convenience for accessing relevant visual memories.

3.4.1 Text-query Image Retrieval. The process of querying the system via text is represented in Figure 2.

For last year’s version, there was ongoing work on using an LLM to determine the entities in a query, keeping only the important terms on a text query. However, for this version, with the addition of image embeddings and retrieval using Milvus and CLIP, we were inspired by the automatic query parser of the LifeInsight 2.0 team at the LSC24 [24] and introduced a query parser for text queries. Similarly, our query parser uses Stanza [19] to identify the main query entities, and we chose DeepSeek-R1-Distill-Qwen-1.5B [5] to extract the main topic of the query. Additionally, each entity can

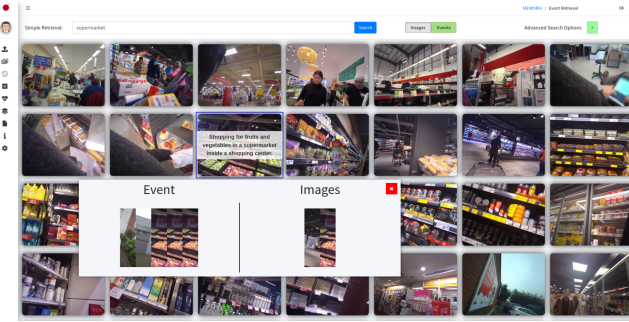


Figure 3: Event Retrieval Page.

be toggled in the user interface to further refine results and allow for a more customized search. The entities extracted by Stanza and filters chosen by the user will be used to search the database of textual annotations while the main topic extracted by the LLM will be used by the embedding model to search the vector database, this functionality can be toggled to use the user query directly. Both results will be interleaved and presented to the user.

3.4.2 Image-to-Image Search Enhancements. Initially, MEMORIA supported image-to-image search by allowing users to upload an image. For such queries, the system generated a textual description of the uploaded image ClipCap and then returned images with highly similar captions. In this iteration, the image-to-image search capability has been significantly enhanced through the integration of vector-based retrieval. MEMORIA now generates image embeddings for the uploaded query image. This approach enables the system to efficiently find and rank the most visually similar images to the query using Milvus, moving beyond text-based caption matching to direct visual feature comparison.

3.4.3 Text-query Event Retrieval. The last way MEMORIA allows for retrieval is by searching for events. With the introduction of the event descriptions described in subsection 3.3, the user can directly search for events using a textual query. This functionality can be used to directly obtain a cluster of similar images that detail an event, and related events that occurred chronologically before or after can be directly explored via the user interface, which is very useful in certain retrieval scenarios.

3.5 Interface

MEMORIA’s interface has been continually enhanced based on user feedback to provide a more powerful and intuitive tool for users to interact with lifelog data. A new page called event retrieval now allows the user to search for an event based on a text query as seen in Figure 3.

Additionally, to help in the competition, some buttons were added in the image pop-up, as seen in Figure 4, described in the previous event’s paper [8], to facilitate and speed up the act of sending the images to the event’s server. The images on the bottom right side of the pop-up can now be selected or deselected to be sent to the server in bulk. One thing to note is that all these images are part of the same event, meaning that they are likely related.

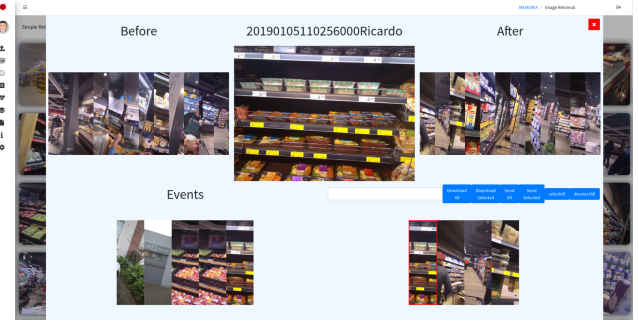


Figure 4: Image Popup page.

3.6 Album feature

A newly introduced feature in MEMORIA is the ability to organize images into user-defined albums. Although this functionality may not directly impact competition tasks, it offers a valuable organizational tool for users to interact with their life-log data more personally. By grouping images into intuitive and meaningful collections, users can structure their memories in ways that reflect their own experiences and preferences, such as collections of similar images or collections of images that belong to a trip. Additionally, the user can also interact with a chatbot powered by the Mistral7B model and a RAG implementation with LLamaIndex that uses the information from all the images in the album to answer questions from the user or describe the content of the album or a portion of it.

4 LSC25 remarks

With the new improvements, the system outperformed its previous versions. The most notable enhancement was the addition of image embeddings and Milvus, as discussed in subsection 3.2. This improvement allowed the team to retrieve relevant lifelogs more quickly and accurately by working with image visual features and text features instead of relying on textual annotations only. The query parser was also useful when Milvus was disabled and the team was using the query entities for retrieval, however, the main topic extraction feature was not as helpful as we had initially thought it would be, the images returned by Milvus with that extra step enabled were almost always the same and sometimes “worse” than what it returned without parsing the query at all.

Overall, MEMORIA was able to rank first place at the end of the LSC25, with very good scores in both QA and KIS topics, and also being ranked first in ADHOC topics. As discussed before, this is largely because of Milvus and vector retrieval. We expect MEMORIA to be more complete next year, mainly with the addition of metadata in Milvus, as it is going to be discussed in Section 5

5 Future Work

For future work, the team aims to add metadata to Milvus with the goal of enabling more refined and efficient filtering during image retrieval. With this enhancement, the already good results provided by Milvus can be filtered to allow for better results, for example when the topic restricts the lifelogs to a certain time frame.

Another planned enhancement is the introduction of conditional querying, which will allow users to combine multiple contextual conditions within a single search. For example, a user could query for images of dining at a restaurant after driving in the rain. The system would first identify time periods when the individual was driving in the rain and then locate subsequent events, such as visits to restaurants, that occurred shortly afterwards.

These improvements aim to make the system more flexible and complete, in order to allow for more accurate retrieval of lifelogs.

6 Conclusion

This paper has presented the current state of the MEMORIA system (Memory Enhancement and MOment Retrleval Application) after participation in the LSC25 challenge. The team is extremely satisfied with the system’s performance, and we expect MEMORIA to be even more performant in the future. We will also analyze the systems that participated this year, as some features presented by them could also be applied to MEMORIA to make it more accurate at retrieval.

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